

Problem : Matrices Commuting with All Pauli Matrices

Problem Statement

Let $\sigma_1, \sigma_2, \sigma_3$ be the Pauli spin matrices. Find all 2×2 matrices A such that

$$[\sigma_1, A] = [\sigma_2, A] = [\sigma_3, A] = 0_2,$$

where $[X, Y] = XY - YX$ denotes the commutator.

Step 1: Recall the Pauli Matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

These matrices form a basis of the space of traceless Hermitian 2×2 matrices and generate the full matrix algebra $M_2(\mathbb{C})$.

Step 2: General Form of a 2×2 Matrix

Any complex 2×2 matrix can be written uniquely as

$$A = a_0 I + a_1 \sigma_1 + a_2 \sigma_2 + a_3 \sigma_3,$$

where $a_0, a_1, a_2, a_3 \in \mathbb{C}$.

Step 3: Use the Commutation Relations

The Pauli matrices satisfy

$$[\sigma_j, \sigma_k] = 2i \sum_{\ell=1}^3 \varepsilon_{j k \ell} \sigma_\ell,$$

where $\varepsilon_{j k \ell}$ is the Levi-Civita symbol.

Compute:

$$[\sigma_j, A] = [\sigma_j, a_0 I] + \sum_{k=1}^3 a_k [\sigma_j, \sigma_k].$$

Since $[\sigma_j, I] = 0$, this becomes

$$[\sigma_j, A] = 2i \sum_{k,\ell=1}^3 a_k \varepsilon_{jk\ell} \sigma_\ell.$$

For this to vanish for all $j = 1, 2, 3$, we must have

$$a_1 = a_2 = a_3 = 0.$$

Step 4: Final Form of A

The only remaining term is:

$$A = a_0 I.$$

Thus,

$$A = \lambda I, \quad \lambda \in \mathbb{C}.$$

Final Answer

The only 2×2 matrices that commute with all Pauli matrices are scalar multiples of the identity.

Key Insight

The Pauli matrices generate the full matrix algebra $M_2(\mathbb{C})$, so their commutant consists only of $\mathbb{C}I$.

This is a special case of *Schur's lemma*, which states that any operator commuting with an irreducible representation is proportional to the identity.

Physical Interpretation

In quantum mechanics, this result implies:

- Any observable commuting with all spin components must be trivial,
- Such an operator carries no directional (spin) information.